

LESSON 2-7 • SESSION 2 • PRACTICE • FORM A

Dividing by a Decimal

Learning target – I can divide by a decimal by making the divisor a whole number first. • 6.NS.B.3

REMEMBER

- Make the divisor whole: multiply **both** numbers by the **same** power of 10.
- Count the divisor's decimal places. One place: $\times 10$. Two places: $\times 100$. Example: $5 \div 0.25 = 500 \div 25 = 20$.
- Dividing by a number less than 1 gives a quotient **greater** than the dividend.

WORKED EXAMPLE

How many 0.7-liter bottles can be filled from 8.4 liters?

Step 1 Write $8.4 \div 0.7$.**Step 2** Multiply both by 10: $0.7 \times 10 = 7$ and $8.4 \times 10 = 84$.**Step 3** $84 \div 7 = 12$. Check: $12 \times 0.7 = 8.4$.**Answer 12 bottles****WORDS** **divisor** (*divisor*) • **equivalent division** (*división equivalente*) • **power of ten** (*potencia de diez*)

PRACTICE

show your work • label units

1. Rewrite with a whole-number divisor. Then divide.

a. $14 \div 0.35 =$ _____

b. $37.2 \div 3.1 =$ _____

c. $12.6 \div 0.9 =$ _____

d. $5 \div 0.25 =$ _____

e. $8.4 \div 0.2 =$ _____

f. $6.5 \div 0.05 =$ _____

- 2.
- Science Lab Pours.**
- A lab pours 7.2 liters of solution into 0.6-liter beakers. Later it pours 4.5 liters into 0.9-liter tubes.

a. Write the whole-number problem. How many beakers? _____

b. Write the whole-number problem. How many tubes? _____

Show your work.

- 3.
- Paint Cans.**
- A painter fills 1.3-liter cans from 11.7 liters of paint. How many cans are filled?

Hint Multiply both numbers by 10.

Show your work.

Answer _____ cans

- 4.
- One-Sided Slide.**
- For
- $6.3 \div 0.7$
- , Alina divided 6.3 by 7. What did her shortcut do to the quotient? Find the correct answer.

Hint Did Alina multiply both numbers by 10?

Show your work.

Quotient _____

- 5.
- Bagging Sugar.**
- A store packs 20.4 kg of sugar into bags that hold 1.2 kg each. How many bags?

Show your work.

Answer _____ bags

- 6.
- Kite Tails.**
- A spool holds 9.1 m of string. Each kite tail uses 0.7 m. How many tails?

Show your work.

Answer _____ tails

- 7. Too Few Slides.** For $3.6 \div 0.12$, Yara moved the divisor's point two places but the dividend's point one place: $36 \div 12$. Explain her mistake and find the correct quotient.

Hint 0.12 has two decimal places, so multiply both by 100.

Show your work.

Quotient _____

- 8. Paint Coverage.** A painter has 12.6 liters of paint. Each door takes 0.6 liter.

Hint *Bigger amount per door means fewer doors.*

- a.** How many doors can the painter cover? _____ doors
- b.** If each door took 0.9 liter, how many doors? _____ doors
- c.** Why does the number of doors drop? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Oren's Work.** For $5.6 \div 0.8$, Oren correctly rewrote the problem as $56 \div 8$, then wrote his answer as 0.7. Explain the mistake and give the correct quotient.

TEST PRACTICE

choose the best answer

- 10.** Find $7.2 \div 0.8$.
- A** 0.9 **B** 9 **C** 90 **D** 5.76
- 11.** To divide by 0.25, you first multiply both numbers by –
- A** 10 **B** 100 **C** 25 **D** 1,000

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

1.3 X	13 M	12 D	17 I	5 E	3 Z	9 C	21 L	30 A	0.9 Q
#2a		#2b	#3	#5	#6	#7	#8a		

CHECK YOURSELF

☐ I can do this on my own
 ☐ I need a little help
 ☐ I have a question about #____